

ECO102 - Introduction to Macroeconomics

Department of Economics and Social Sciences (ESS), Brac University

Measuring GDP and Economic Growth

Lecture 1

Reading

- Economics by Michael Parkin, 10th edition
 - Chapter: 21

Outline

- Define GDP and use the circular flow model to explain why GDP equals aggregate expenditure and aggregate income
- Explain how the Bureau of Economic Analysis measures U.S. GDP and real GDP
- Describe how real GDP is used to measure economic growth and fluctuations and explain the limitations of real GDP as a measure of economic well-being

Introduction

- What exactly is GDP and what does it tell us about the state of the economy?
- Some countries are rich while others are poor. How do we compare economic well-being in one country with that in another?

Gross Domestic Product

GDP, or gross domestic product, is the market value of the final goods and services produced within a country in a given time period. This definition has four parts:

- Market value
- Final goods and services
- Produced within a country
- In a given time period

Market Value

- GDP values items produced in an economy at their market values – the prices at which items are traded in markets
- For example, if the price of an apple is 10 cents, then the market value of 50 apples is \$5 and if the price of an orange is 20 cents, then the market value of 100 oranges is \$20
- By using market prices to value production, we can add the apples and oranges together. The market value of 50 apples and 100 oranges is \$5 plus \$20, or \$25

Final Good and Services

- To calculate GDP, we value the final goods and services produced
- Final good vs intermediate good – A Dell computer is a final good, but an Intel Pentium chip inside it is an intermediate good
- Some goods can be an intermediate good in some situations and a final good in other situations. For example, ice cream we buy vs the ice cream restaurant buys to make sundae
- Some items that people buy are neither final goods nor intermediate goods and they are not part of GDP – A secondhand good was part of GDP in the year in which it was produced, but not in GDP this year (e.g used cars, existing homes, stocks and bonds)

Produced Within a Country & In a Given Time Period

❑ Produced Within a Country

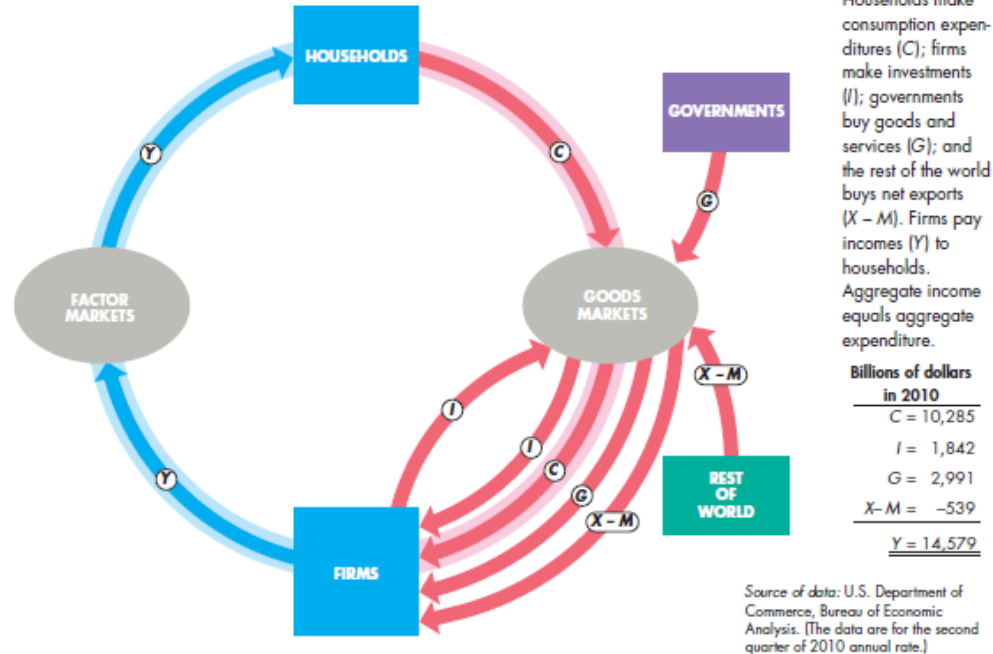
- Only goods and services that are produced within a country count as part of that country's GDP
- Nike Corporation, a U.S. firm, produces sneakers in Vietnam, and the market value of those shoes is part of Vietnam's GDP, not part of U.S. GDP

❑ In a Given Time Period

- GDP measures the value of production in a given time period—normally either a quarter of a year—called the quarterly GDP data—or a year—called the annual GDP data

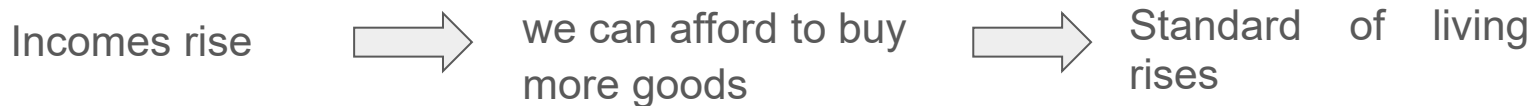
GDP and the Circular Flow of Expenditure and Income

FIGURE 21.1 The Circular Flow of Expenditure and Income



GDP and the Circular Flow of Expenditure and Income

- GDP measures not only the value of total production but also total income and total expenditure
- The equality between the value of total production and total income shows the direct link between productivity and living standards



- But we must produce more goods and services if we are to be able to buy more goods and services
- Rising incomes and a rising value of production are two aspects of the same phenomenon: increasing productivity

GDP Equals Expenditure Equals Income

- Gross domestic product can be measured in two ways: By the **total expenditure** on goods and services or by the **total income** earned producing goods and services

$$Y = C + I + G + X - M$$

- GDP equals aggregate expenditure and equals aggregate income

Measuring GDP - Expenditure Approach

- $C \Rightarrow$ Personal consumption expenditures (e.g books, banking services, bread, TVs)
- $I \Rightarrow$ Gross private domestic investment (e.g expenditure on capital equipment, inventories and buildings by firms; expenditure on new homes by households)
- $G \Rightarrow$ Government expenditure on goods and services
(e.g expenditure on national defense)

Note: It does not include transfer payments, such as unemployment benefits

- $X - M \Rightarrow$ Net exports of goods and services (value of exports minus the value of imports)

Measuring GDP - Expenditure Approach

TABLE 21.1 GDP: The Expenditure Approach

Item	Symbol	Amount in 2010 (billions of dollars)	Percentage of GDP
Personal consumption expenditures	C	10,285	70.5
Gross private domestic investment	I	1,842	12.6
Government expenditure on goods and services	G	2,991	20.5
Net exports of goods and services	$X - M$	-539	-3.7
Gross domestic product	Y	14,579	100.0

Measuring GDP - Income Approach

- The income approach measures GDP by summing the incomes that firms pay households for the services of the factors of production they hire—wages for labor, interest for capital, rent for land, and profit for entrepreneurship
- The National Income and Product Accounts divide incomes into two big categories:
 1. Compensation of employees – net wages and salaries (called “take-home pay”) that workers receive plus taxes withheld on earnings plus fringe benefits such as Social Security and pension fund contributions
 2. Net operating surplus – the sum of all other factor incomes (net interest, rental income, corporate profits, and proprietors’ income)

Measuring GDP - Income Approach

TABLE 21.2 GDP: The Income Approach

Item	Amount in 2010 (billions of dollars)	Percentage of GDP
Compensation of employees	7,929	54.4
Net interest	924	6.3
Rental income	299	2.1
Corporate profits	1,210	8.3
Proprietors' income	<u>1,050</u>	<u>7.2</u>
<i>Net domestic income at factor cost</i>	11,412	78.3
Indirect taxes <i>less</i> subsidies	<u>1,127</u>	<u>7.7</u>
<i>Net domestic income at market prices</i>	12,539	86.0
Depreciation	<u>1,860</u>	<u>12.8</u>
GDP (income approach)	14,399	98.8
Statistical discrepancy	<u>180</u>	<u>1.2</u>
GDP (expenditure approach)	<u>14,579</u>	<u>100.0</u>

Nominal GDP and Real GDP

- **Nominal GDP** is the value of final goods and services produced in a given year when valued at the prices of that year. Nominal GDP is just a more precise name for GDP
- **Real GDP** is the value of final goods and services produced in a given year when valued at the prices of a reference base year. By comparing the value of production in the two years at the same prices, we reveal the change in production

Nominal GDP and Real GDP

TABLE 21.3 Calculating Nominal GDP and Real GDP

		Quantity (millions)	Price (dollars)	Expenditure (millions of dollars)
Item				
(a) In 2005				
C	T-shirts	10	5	50
I	Computer chips	3	10	30
G	Security services	1	20	20
Y	Real and Nominal GDP in 2005			100
(b) In 2010				
C	T-shirts	4	5	20
I	Computer chips	2	20	40
G	Security services	6	40	240
Y	Nominal GDP in 2010			300
(c) Quantities of 2010 valued at prices of 2005				
C	T-shirts			
I	Computer chips			
G	Security services			
Y	Real GDP in 2010			

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Y	Nominal GDP in 2010			300
(c) Quantities of 2010 valued at prices of 2005				
C	T-shirts	4	5	20
I	Computer chips	2	10	20
G	Security services	6	20	120
Y	Real GDP in 2010			160

Nominal GDP in 2010 is
three times its value in
2005

Real GDP in 2010 is 1.6
times its 2005 value –
60% increase in
production

Uses of Real GDP

Economists use estimates of real GDP for two main purposes:

- To compare the standard of living over time
- To compare the standard of living across countries

❑ Compare Standard of Living over time

Long-term Trend

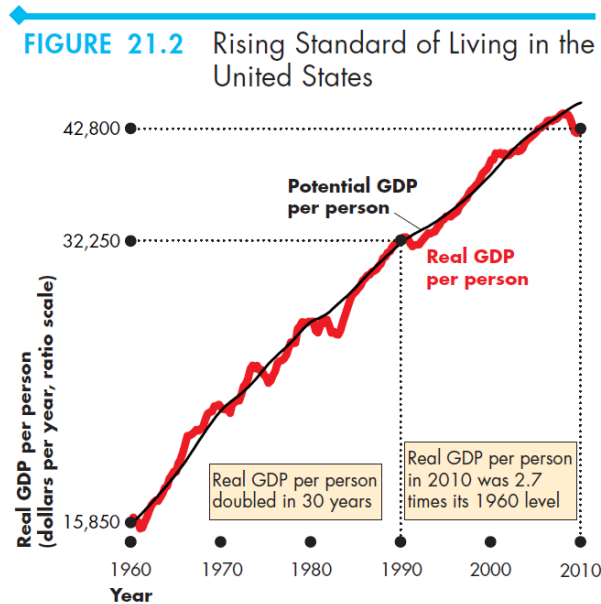
- **Real GDP per person** tells us the value of goods and services that the average person can enjoy

$$\text{Real GDP person} = \text{Real GDP} \div \text{Population}$$

- By using real GDP, we remove any influence that rising prices and a rising cost of living might have had on our comparison
- Real GDP person overtime is expressed as a ratio of some reference year
- For example, in 1960, real GDP per person was \$15,850 and in 2010, it was \$42,800. So real GDP per person in 2010 was $(\$42,800 \div \$15,850)$ 2.7 times its 1960 level

❑ Compare Standard of Living over time

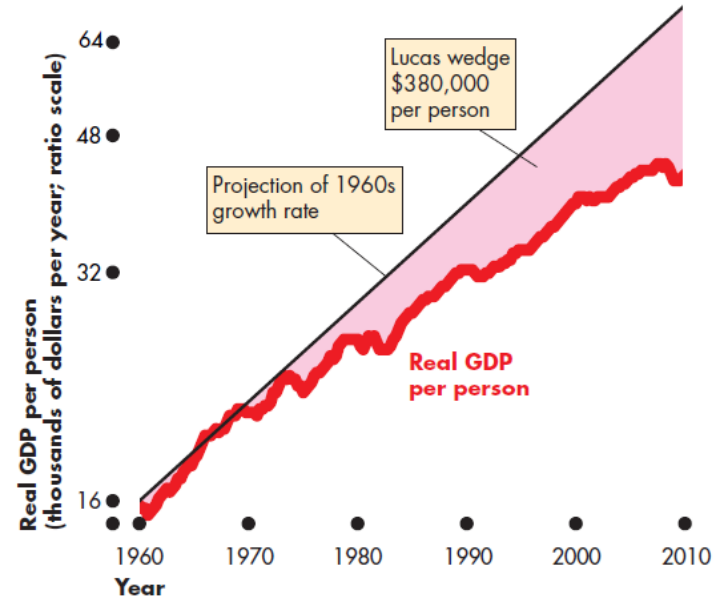
The Growth of Potential GDP and Fluctuations of Real GDP



Potential GDP is the maximum level of real GDP that can be produced while avoiding shortages of labor, capital, land, and entrepreneurial ability that would bring rising inflation

❑ Compare Standard of Living over time

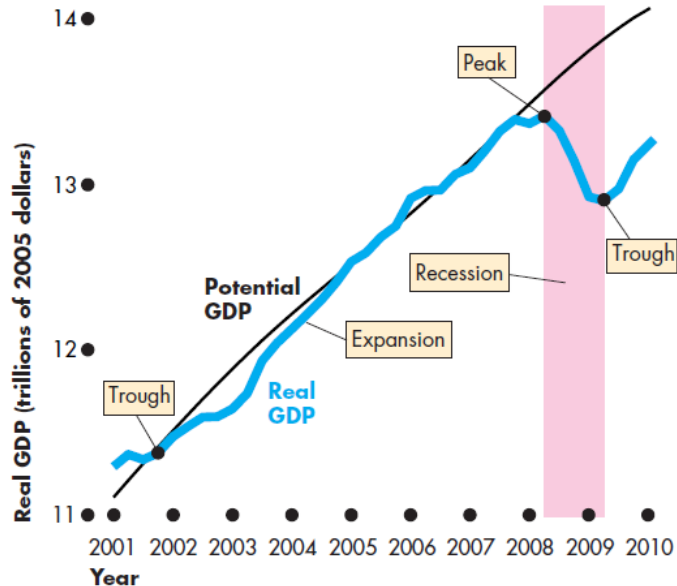
FIGURE 21.3 The Cost of Slower Growth:
The Lucas Wedge



❑ Compare Standard of Living over time

Real GDP Fluctuations – The Business Cycle

FIGURE 21.4 The Most Recent U.S. Business Cycle



- An expansion is a period during which real GDP increases
- Recession is a period during which real GDP decreases — its growth rate is negative — for at least two successive quarters
- NBER define recession as “a period of significant decline in total output, income, employment, and trade, usually lasting from six months to a year, and marked by contractions in many sectors of the economy”

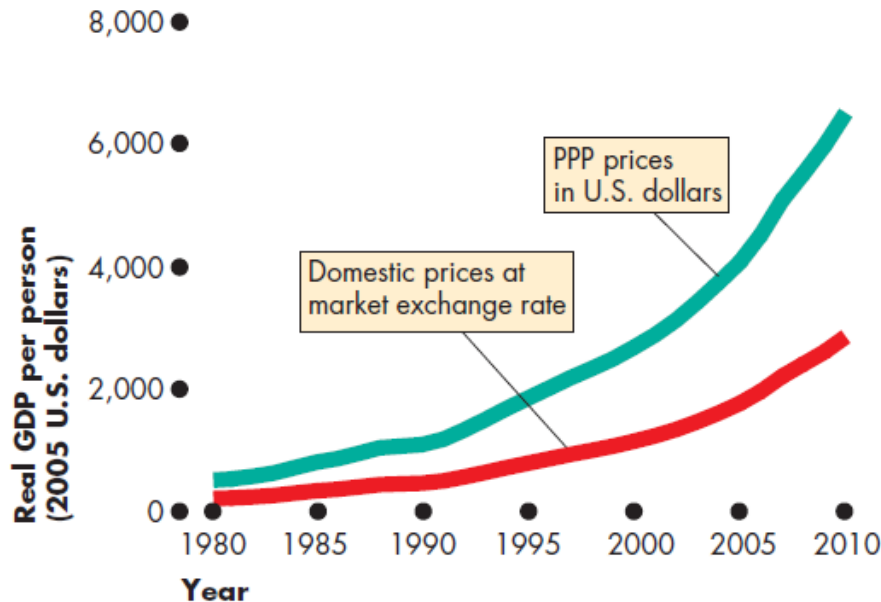
❑ Compare Standard of Living across Countries

To compare living standards between US and China, what problems arise?

- First, the real GDP of one country must be converted into the same currency units as the real GDP of the other country
- Second, the goods and services in both countries must be valued at the same prices

❑ Compare Standard of Living across Countries

FIGURE 21.5 Two Views of Real GDP in China



- When GDP in 2010 is valued at the market exchange rate, U.S. income per person is 15 times that in China
- When GDP is valued at purchasing power parity prices, U.S. income per person is only 6.5 times that in China

Limitations of Real GDP

Real GDP measures the value of goods and services that are bought in markets. Some of the factors that influence the standard of living and that are not part of GDP are:

- Household production
- Underground economic activity
- Health and life expectancy
- Leisure time
- Environmental quality
- Political freedom and social justice

Limitations of Real GDP

■ Household production:

The omission of household production from GDP means that GDP underestimates total production. But it also means that the growth rate of GDP overestimates the growth rate of total production – the increase in GDP arises from a decrease in home production

■ Underground economic activity

Underground economic activity is unreported, it is omitted from GDP

■ Health and life expectancy

- Good health and a long life do not show up in real GDP, at least not directly
- New health and life expectancy problems every year, when taken into account, we see that real GDP growth overstates the improvements in the standard of living

Limitations of Real GDP

■ Leisure time

Leisure time is an economic good that adds to our economic well-being and the standard of living. However, working time is valued as part of GDP, but leisure time is not reflected in real GDP

■ Environmental quality

Economic activity directly influences the quality of the environment – but deteriorating air quality not counted as negative part of GDP

■ Political freedom and social justice

A country might have a very large real GDP per person but have limited political freedom and social justice

GDP per capita growth (annual %) - Bangladesh

